

The statistics of soaring flight

EconophysicsLab offsite 2026



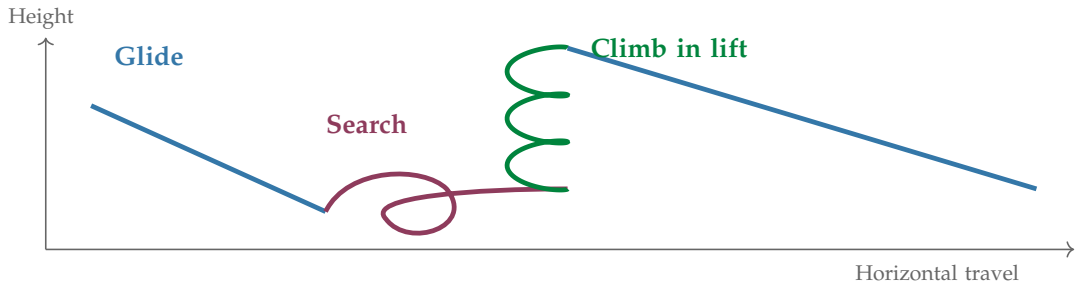
Matteo Di Sante

Transport, routes and
launch environments

Soaring is transport with an energy budget

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Height is spent to travel and recovered in rising air.



Wind, terrain and route choice act throughout the flight.

Schematic

Which statistical laws emerge from this sequence of behaviours?

A large archive of free flight

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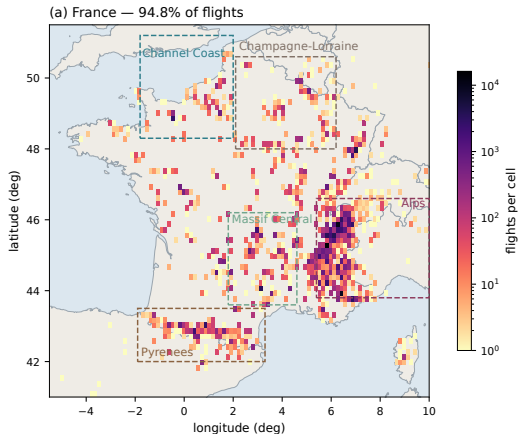
FFVL distance competition

186,052

Paraglider tracklogs
1999–2000 to 2025–2026

6,716

Hang-glider tracklogs
2001–2002 to 2025–2026



IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

Preprocessing: clean the recorded flight

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1 Altitude channel

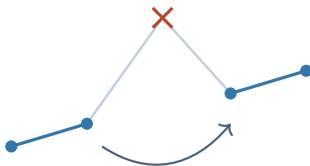
Use GNSS altitude throughout.
Check presence and range.



One altitude channel

2 Fix-level cleaning

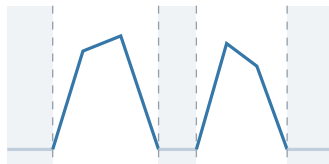
Remove impossible positions.
Invalidate bad altitude fixes.



Require a successful rejoin

3 Airborne interval

Trim ground phases.
Split witnessed ground stops.



Trim ends; split ground stops

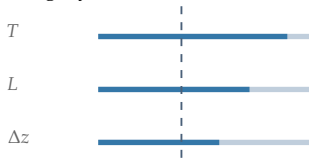
An outlier flag alone never deletes a fix; physical checks decide removal

Preprocessing: select and standardise trajectories

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4 Flight selection

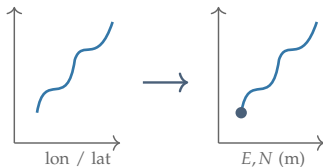
Duration ≥ 40 min.
Path ≥ 20 km.
Altitude range ≥ 600 m.
Integrity and reach checks.



Apply flight thresholds

5 Coordinates

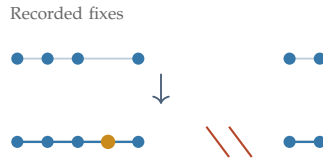
Project to local east and north.
Keep one origin per flight.



One local coordinate frame

6 Time grid

Fill short gaps; split long gaps.
Flag repairs, then smooth
and differentiate.



Short gap filled; long gap split

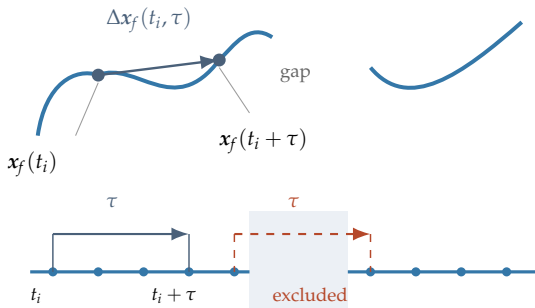
156,406 paragliders 84.1%

6,094 hang gliders 90.7%

Measuring horizontal transport

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One flight f , two retained segments



$\mathcal{I}_f(\tau)$: origin indices with both endpoints inside the same retained segment.

Pool origins within each flight;
then give each flight equal weight.

$$\Delta \mathbf{x}_f(t_i, \tau) = \mathbf{x}_f(t_i + \tau) - \mathbf{x}_f(t_i)$$

$$m_f(\tau) = \frac{1}{|\mathcal{I}_f(\tau)|} \sum_{i \in \mathcal{I}_f(\tau)} |\Delta \mathbf{x}_f(t_i, \tau)|^2$$

$$M_2(\tau) = \frac{1}{|\mathcal{C}(\tau)|} \sum_{f \in \mathcal{C}(\tau)} m_f(\tau)$$

$\mathcal{C}(\tau) = \{f : |\mathcal{I}_f(\tau)| > 0\}$: contributing flights.

$$M_2(\tau) \propto \tau^{2H}: \quad H = \frac{1}{2} \text{ diffusion} \quad H = 1 \text{ ballistic motion}$$

Keep the same flights at every lag

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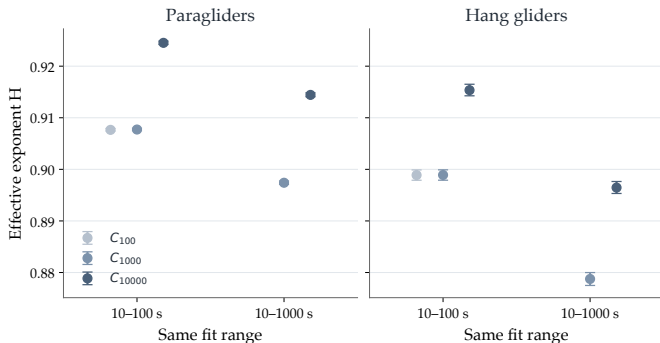
One cohort for 10–10,000 s

$$\tau_{\max} \leq 0.8T_s$$

Selected segments last
at least 12,500 s (3 h 28 min).

46,273 paragliders

2,326 hang gliders



Duration selection raises H by about 2% on matched lag ranges.

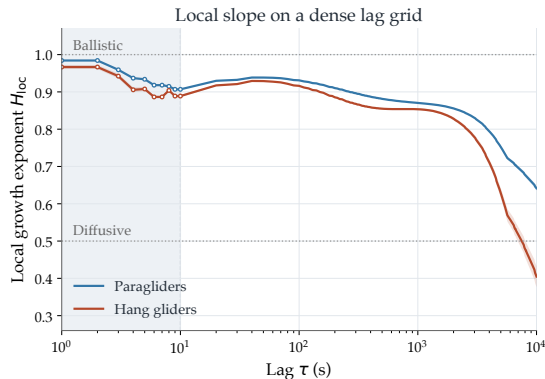
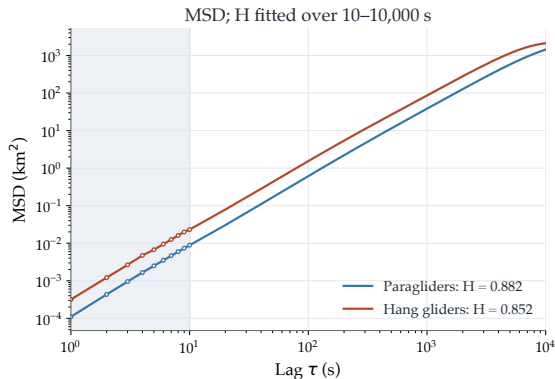
A fixed sample makes MSDs comparable across lags: 10–10,000 s

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Displacement growth from 1 s to 10,000 s

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C_{10000} : flights enter as the lag reaches their logging interval.



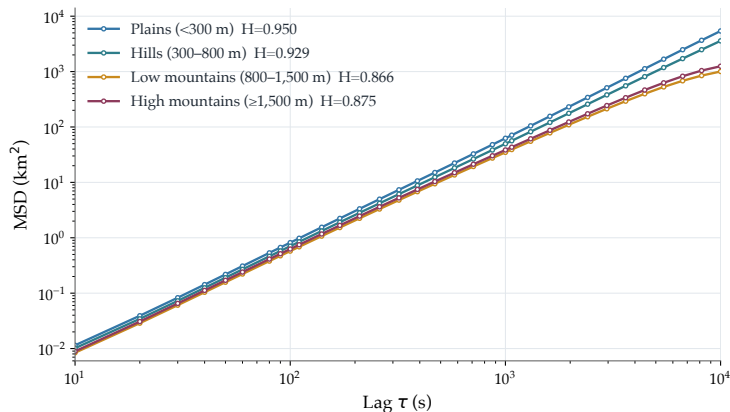
Shaded range: the sample changes, so the local slope also reflects composition.

Full cohort from 10 s: 46,273 paragliders and 2,326 hang gliders

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Launch-altitude groups have different MSDs

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Circuit composition

Open: no return required.

Closed: a return circuit.

	Open	Closed
Plains	91.1%	8.9%
Hills	71.1%	28.9%
Low mtn.	33.5%	66.3%
High mtn.	35.0%	64.8%

Shares within each altitude group.

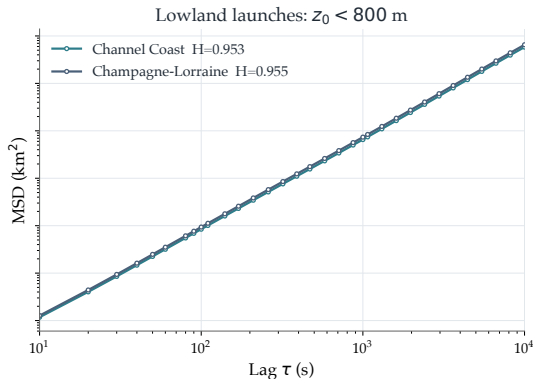
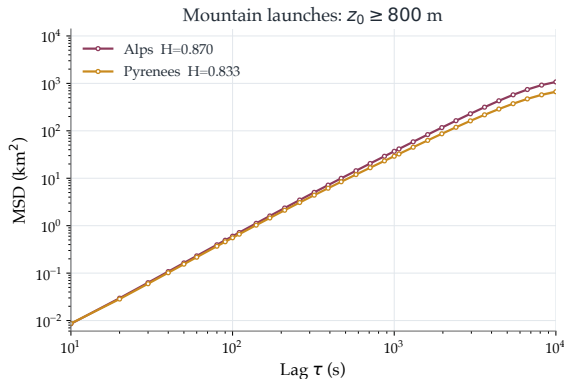
Unclassified: less than 0.3%.

Plains contain more open circuits, associated with higher H (slide 12).

The pooled altitude contrast combines terrain setting and circuit composition.

Region matters within a broad launch environment

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Alps–Pyrenees: $\Delta H = 0.037$

Lowland pair: $\Delta H = -0.002$, interval includes zero

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Viewer: intersect climb-labelled trajectories with horizontal planes.

Within each cell: same dates, segmentation and map bounds at both heights.

Mountains

Plains

$$z_{AGL} = z_1$$

$$z_{AGL} = z_2$$

$$z_{AGL} = z_1$$

$$z_{AGL} = z_2$$

Mountain cell
lower plane

Viewer screenshot

Mountain cell
higher plane

Viewer screenshot

Plains cell
lower plane

Viewer screenshot

Plains cell
higher plane

Viewer screenshot

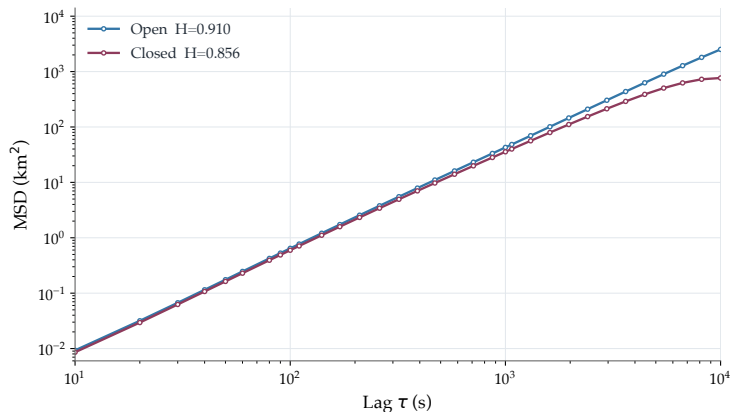
Working observation: mountain patterns shift less with height than lowland patterns.

Possible explanations: stronger uplift or weaker wind; neither is measured here.

Dots are climb crossings, not thermal centres. z_{AGL} is relative to a fixed reference for each cell.

Open routes have larger displacement growth

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Open circuits



17,961 flights

Closed circuits



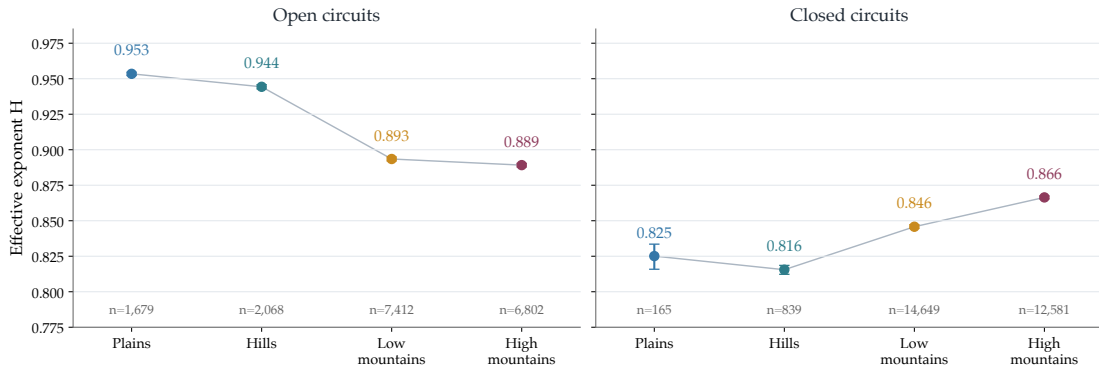
28,234 flights

$$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] \quad (90\% \text{ paired interval})$$

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

The altitude ordering reverses for closed routes

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Interpretation: terrain-linked lift may constrain mountain routes, limiting both straight progress and direct returns to close a circuit.

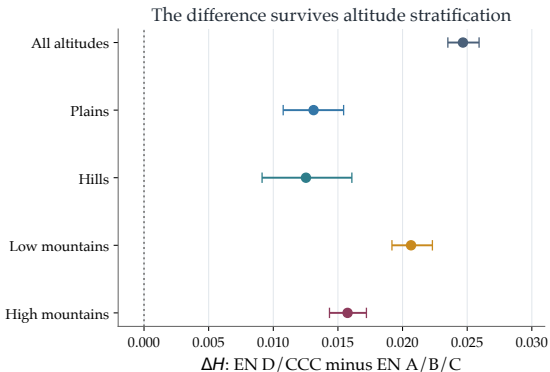
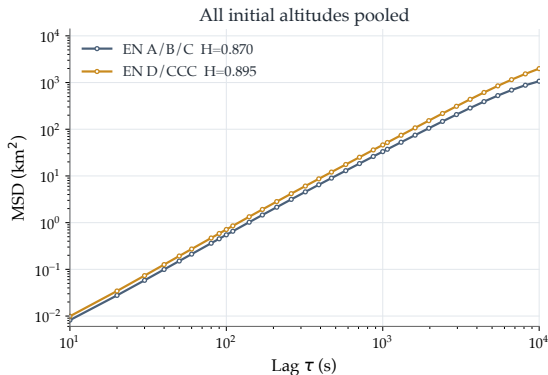
Open: lowland H is higher

Closed: mountain H is higher

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

The equipment contrast survives altitude stratification

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Higher H for EN D/CCC in every altitude class; largest gap in Low mountains

Wing class is a proxy. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

1 Group flights

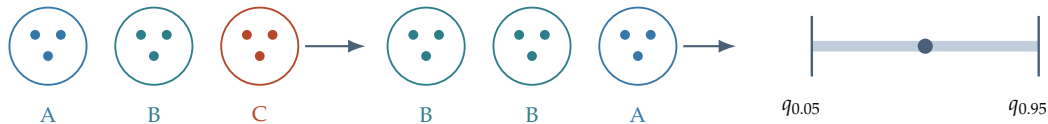
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined